DigiPass

Know Your Products!

From Bricks to Forms

Building Digital Product Passports with SHACL

A visual approach to structured data

No coding background required

COORDINATION AND KNOWLEDGE SHARING ACROSS MATERIALS DEVELOPMENT COMMUNITIES (CSA):
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WIKKI LIMITED, UK participant in Horizon Europe Project DigiPass, is supported by UKRI grant number 10100819: **DigiPass**

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Digital Product Passports Are Hard

❌ **TODAY**

- Company A says: 'weight: 5kg'
- Company B says: 'mass: 5000g'
- Company C says: '5' (just a number!)

→ Computers can't understand this
→ Manual data entry & errors
→ Compliance headaches

✓ **WITH SHACL**

- Product.weight = 5
- Must be a number
- Must have a unit
- Unit can only be: kg, g, lb, oz

→ Machines can validate it
→ Forms guide data entry
→ Always consistent

What If Data Had Grammar?

Like grammar rules in language, data needs rules too.

❌ "The product weighs about 5kg usually" (ambiguous)

```
✓   Product.weight  =  5
         type    →  number
         unit    →  kg  (or g, lb, oz)
     required    →  yes
      checked    →  automatically
```

SHACL is the W3C standard for writing these rules.

Lego Bricks for Data Schemas

Instead of writing rules from scratch every time → create reusable bricks once, snap them together.

🧱 Mass

number + unit
(kg, g, lb, oz, mg, t)

🌡️ Temperature

number + degrees
(°C, °F, K)

⚡ Battery Cap

number + unit
(Ah, mAh, Coulombs)

🏭 Manufacturer

name + ID + country
(linked data)

Define once → reuse across every product schema in your organisation.

Creating a Reusable Brick

Point-and-click — no coding needed:

Name	BatteryCapacity
Path	ex:batteryCapacity
Datatype	xsd:decimal
Class ✨	qudt:ElectricCharge ← triggers magic

Result →

```
Unit dropdown appears
automatically:

 ▼  Ampere-hour (Ah)
    Milliampere-hour (mAh)
    Coulomb (C)
    Kiloampere-hour (kAh)
```

The system looks up applicable units from the QUDT ontology — no manual list needed.

Assembling Bricks into a Product Schema

```
EV Battery Schema
├── Battery Identification
│   ├── Model Number        (text)
│   ├── Serial Number       (text)
│   └── Manufacturer        (linked data)
├── Physical Properties
│   ├── Mass            ← [Mass brick]          → unit dropdown
│   └── Capacity        ← [BatteryCapacity]     → unit dropdown
└── Environmental
    ├── Carbon Footprint   ← [CO₂ brick]         → unit dropdown
    └── Recyclability      ← [Percent brick]     → % field
```

Visual tree editor — drag bricks in, set connections, done.

One Source → Many Outputs

The Magic: One Brick, Three Things

Brick (JSON)

```
{
  name: 'Mass',
  datatype: xsd:decimal,
  class: qudt:Mass,
  units: [kg,g,lb,...]
}
```

SHACL Rules

```
sh:property [
  sh:path ex:mass ;
  sh:datatype xsd:decimal ;
  sh:class qudt:Mass ;
  sh:in (kg g lb oz) ;
] .
```

User Form

```
Mass
| 5 | ▼ kg |

Change brick once
→ all forms update
```

Smart Widgets from Context

1. You type: qudt:Mass

↓

2. System queries QUDT ontology

↓

3. Finds 6 applicable units

↓

4. Creates unit dropdown automatically

Works for any physical quantity: temperature, pressure, energy, voltage …

Architecture

How It All Connects

Shared Ontologies	QUDT · Schema.org · FOAF · SKOS (17 vocabularies)
Brick Library	Reusable building blocks (JSON)
Schema Composer	Assemble bricks into product blueprints
SHACL Exporter	Generates validation rules (Turtle)
Form Generator	HTML interface the data-entry clerk sees
Validated Data	Ready for Digital Product Passports

Why This Matters

	Before	After
Schema definition time	3 months	1 week
Unit configuration	manual lookup	automatic
Validation errors	50+ (found late)	2 (caught early)
Consistency	varies per team	standardised
Non-technical authors	impossible	point-and-click

Key Takaways

🧱 **Bricks**	Reusable, semantic data components — define once, use everywhere
🔗 **SHACL**	Standardised validation rules — machines can check your data
✨ **Enrichment**	Ontology-driven smart widgets — unit dropdowns appear automatically
🎨 **Forms**	Automatic user interfaces — no coding needed for data entry
♻️ **Reuse**	Change a brick once ⟶ every form using it updates

Getting Started

One Download, One Command

1. **Download** — Clone or download the repository from GitHub
2. **Run** — Clone or download the repository from GitHub
3. **Open** — Clone or download the repository from GitHub
4. **Your data** — Clone or download the repository from GitHub

Questions?

Q: Can non-programmers use this?

A: Yes — brick_app and schema_app are point-and-click web tools.

Q: What about our own vocabulary?

A: Load any RDF/OWL ontology — enrichment works with it automatically.

Q: Does it connect to our database?

A: Exported SHACL validates data; can export to JSON/XML/RDF for any system.

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